Launch Control XL 3 — Synth Board (sounds/lab)

One bass voice; each column is one stage of the signal path: source → volume envelope → filter → filter envelope → FM → pitch → drive → space.

Button row 1 picks a choice within the stage, button row 2 switches the stage on/off (A/B comparison). Exception column 1: button 2 = bassline. At start only envelope and filter are on.

The status box at the bottom right shows all values live, with units. Faders jump from their start value to their real position on first movement.

Knob Fader Button unused

Mode 16 — Synth Board

	1 Source	2 Envelope	3 Filter	4 Filter env.	5 FM	6 Pitch	7 Drive	8 Space
Row 1	CC 13 **Pulse width** Only for pulse: 0.5 = square (hollow) … 0.05 = thin, nasal.	CC 14 **Attack** 1 ms–1 s: how long the note takes to fade in. Short = click, long = soft, swelling.	CC 15 **Resonance** 0–20: boosts the frequencies around the cutoff. High = whistling, squeaky (acid).	CC 16 **Filter attack** 1 ms–1 s: how fast the filter opens. Long = „wah“.	CC 17 **FM decay** 10 ms–2 s: how fast the FM overtones fade (with FM sustain below 1).	CC 18 **Pitch decay** 10 ms–1 s: how fast the pitch falls.	CC 19 **Level after** 0.1–1: distortion makes it louder; use this to bring it back down.	CC 20 **Room size** 0.5–8: small room … cathedral.
Row 2	CC 21 **Detune** Only for supersaw: how far the 5 sawtooths are detuned. 0 = one saw, more = wide, shimmering.	CC 22 **Decay** 10 ms–2 s: how fast it then falls to the sustain level.	CC 23 **HPF** 20 Hz–2 kHz: high-pass, removes the body; the bass gets thin.	CC 24 **Filter decay** 10 ms–2 s: how fast it closes again. Short = „blubb“, squelch.	CC 25 **FM sustain** 0–1: 1 = FM stays for the whole note, 0 = only a strike (like e-piano, bell).	CC 26 **Vibrato rate** 0–12 Hz: how fast the pitch wobbles.	CC 27 **Bitcrush** 16–2 bit: coarser volume steps, rough, digital, 8-bit.	CC 28 **Delay** 0–0.8: volume of the echoes.
Row 3	CC 29 **Noise** 0–1, mix in noise (sine/ triangle/sawtooth/square): air, bite.	CC 30 **Release** 10 ms–2 s: fade-out after the note ends. Mostly audible with short note lengths.	CC 31 **HPF resonance** 0–20: resonance of the high-pass.	CC 32 **Filter sustain** 0–1: how far open it stays while the note sounds.	CC 33 **Detune** 0–5 % off the ratio: crooked, inharmonic, beating.	CC 34 **Vibrato depth** 0–2 semitones: how far it wobbles.	CC 35 **Coarse** 1–32: lower sample rate, metallic aliasing, telephone/Game Boy.	CC 36 **Feedback** 0–0.9: how many echoes (high = endless).
Fader Level	CC 5 **Level** 0–1, overall volume.	CC 6 **Sustain** 0–1: level while the note is held. 0 = every note dies away (pluck), 1 = organ.	CC 7 **Cutoff (LPF)** 40 Hz–12 kHz: low-pass, low = dull, high = bright. Almost no effect on sine (no overtones).	CC 8 **Env amount** 0–6 octaves: how far the filter sweeps up (or down) from the cutoff on each note. Needs column 3 on.	CC 9 **FM amount** 0–8: 0 = none, a little = warmer, a lot = metallic, bell-like, growl.	CC 10 **Pitch env** 0–24 semitones: each note starts this much higher and falls down. High + short = „zap“, kick-like.	CC 11 **Drive** 0–6: how much distortion. The type (button 1) sets the character.	CC 12 **Reverb** 0–1: amount of reverb. Use sparingly on bass, otherwise it turns to mush.
Buttons 1	CC 37 **Waveform** sine, triangle, saw, square, pulse, supersaw.	CC 38 **Note length** ¼, ½, full, ×2 of the step.	CC 39 **Filter type** 12 dB, 24 dB, ladder (Moog).	CC 40 **Direction** up / down per note.	CC 41 **Ratio** 1, 2, 3, 0.5, 1.414 (bell-like).	CC 42 **Octave** 0, +1, +2. Laptop: +1.	CC 43 **Distortion type** 9 types: scurve … chebyshev.	CC 44 **Delay time** 1/8, 3/16, 1/4, 1/16.
Buttons 2	CC 45 **Bassline** Offbeat → Rolling → Octaves → Syncopated.	CC 46 **Envelope on/off** Off = hard on/off switch (organ).	CC 47 **Filter on/off**	CC 48 **Filter env on/off** Only works with filter on.	CC 49 **FM on/off**	CC 50 **Pitch/vibrato on/off**	CC 51 **Drive on/off**	CC 52 **Space on/off**